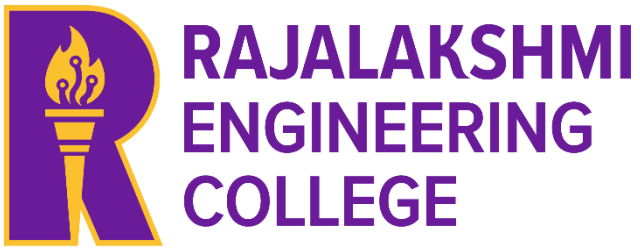
RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

Name :	:	PRAKASH A
Year / Branch / Section	:	II / BE Computer Science and Design / -
University Register No.	:	2116251701041
College Roll No. :	:	251701041
Semester:	:	III
Academic Year :	:	2026 - 2027

FINDING TIME COMPLEXITY OF ALGORITHMS

EXPERIMENT 01

Problem 1: Finding Complexity using Counter Method

Finding Time Complexity of Algorithms

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Sunday, 9 August 2026, 12:16 PM
State	Finished
Completed on	Sunday, 9 August 2026, 12:40 PM
Time taken	23 mins 13 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)

{
    int i= 1;
    int s =1;
    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Reset answer

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
#include<stdio.h>

int main()
{
    int n;
    int c=0;
    int i=1;
    int s=1;
    scanf("%d",&n);
    while(s<n)
    {
        i++;
        s+=i;
        c+=3;
    }
    c+=3;
    printf("%d",c);
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 02

Problem 2: Finding Complexity using Counter method

Finding Time Complexity of Algorithms

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Monday, 3 August 2026, 3:37 PM
State	Finished
Completed on	Monday, 3 August 2026, 3:48 PM
Time taken	11 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

Reset answer

```c
#include<stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    int c=0;
    c++;
    if(n==1)
    {
        //printf("*");
        c++;
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            c++;
            for(int j=1; j<=n; j++)
            {
                c++;
                //printf("*");
                c++;
                //printf("*");
                c++;
                break;

            }
            c++;

        }
        c++;

    }
    printf("%d",c);
}

```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 03

Problem 3: Finding Complexity using Counter Method

Finding Time Complexity of Algorithms

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Monday, 3 August 2026, 3:50 PM
State	Finished
Completed on	Monday, 3 August 2026, 3:58 PM
Time taken	8 mins 26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```c
#include<stdio.h>
int main()
{
    int num;
    int c=0;
    scanf("%d",&num);
    for (int i = 1; i <= num;++i)
    {
        c++;
        if (num % i== 0)
        {
            c++;
            //printf("%d ", i);
        }
        c++;

    }
    c++;
    printf("%d",c);
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

EXPERIMENT 04

Problem 4: Finding Complexity using Counter Method

Finding Time Complexity of Algorithms

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Tuesday, 25 August 2026, 2:56 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 3:32 PM
Time taken	36 mins 18 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```c
#include<stdio.h>
int main()
{
    int c= 1,n;
    scanf("%d",&n);
    for(int i=n/2; i<n; i++)
    {
        c++;
        for(int j=1; j<n; j = 2 * j)
        {
            c++;
            for(int k=1; k<n; k = k * 2)
            {
                c++;
                c++;
            }
            c++;
        }
        c++;
    }
    c++;
    printf("%d",c);
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Problem 5: Finding Complexity using counter method

Finding Time Complexity of Algorithms

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Wednesday, 26 August 2026, 10:43 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 10:46 PM
Time taken	3 mins 3 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```c
#include<stdio.h>
int main()
{
    int n,c=1;
    scanf("%d",&n);
    int rev = 0, remainder;
    c++;
    while (n != 0)
    {
        c++;
        remainder = n % 10;
        c++;
        rev = rev * 10 + remainder;
        c++;
        n/= 10;
        c++;

    }
        c++;
    printf("%d",c);
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.